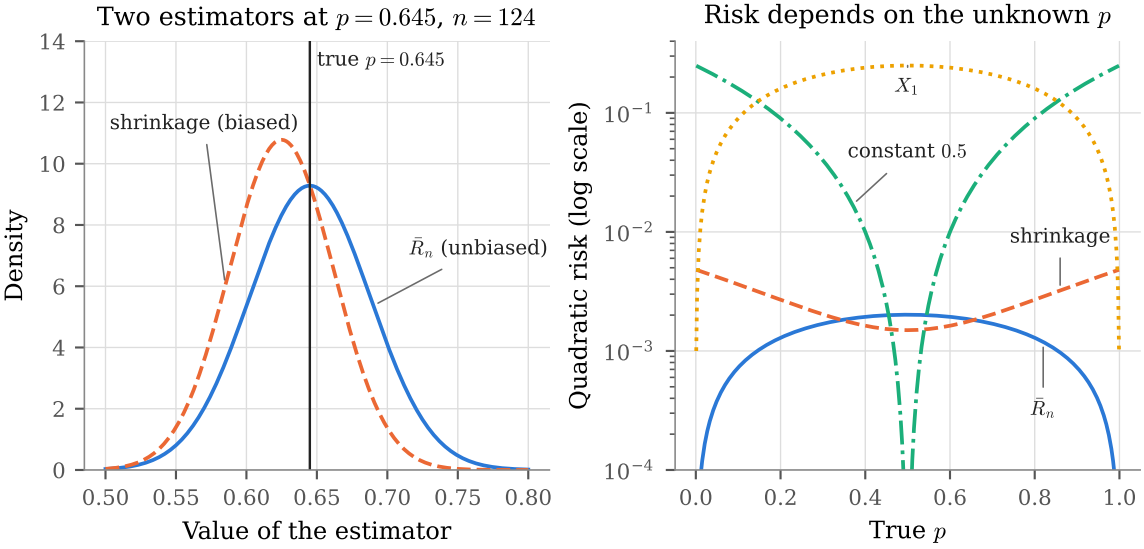
Two estimators at $p = 0.645$, $n = 124$
true $p = 0.645$
shrinkage (biased)
$\bar{R}_n$ (unbiased)
Density
Value of the estimator
Risk depends on the unknown p
X_1
constant 0.5
shrinkage
$\bar{R}_n$
Quadratic risk (log scale)
True p